Ansys
2025 R2

Ansys Fluent Simulation Report

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Date	2/8/2026 10:26 AM

Table of Contents

System Information

Application	Fluent
Settings	2d, double precision, density-based explicit, laminar
Version	25.2.0-10204
Source Revision	5eecd5d865
Build Time	Jun 16 2025 10:44:41 EDT
CPU	AMD Ryzen 7 6800H with Radeon Graphics
OS	Windows

Geometry and Mesh

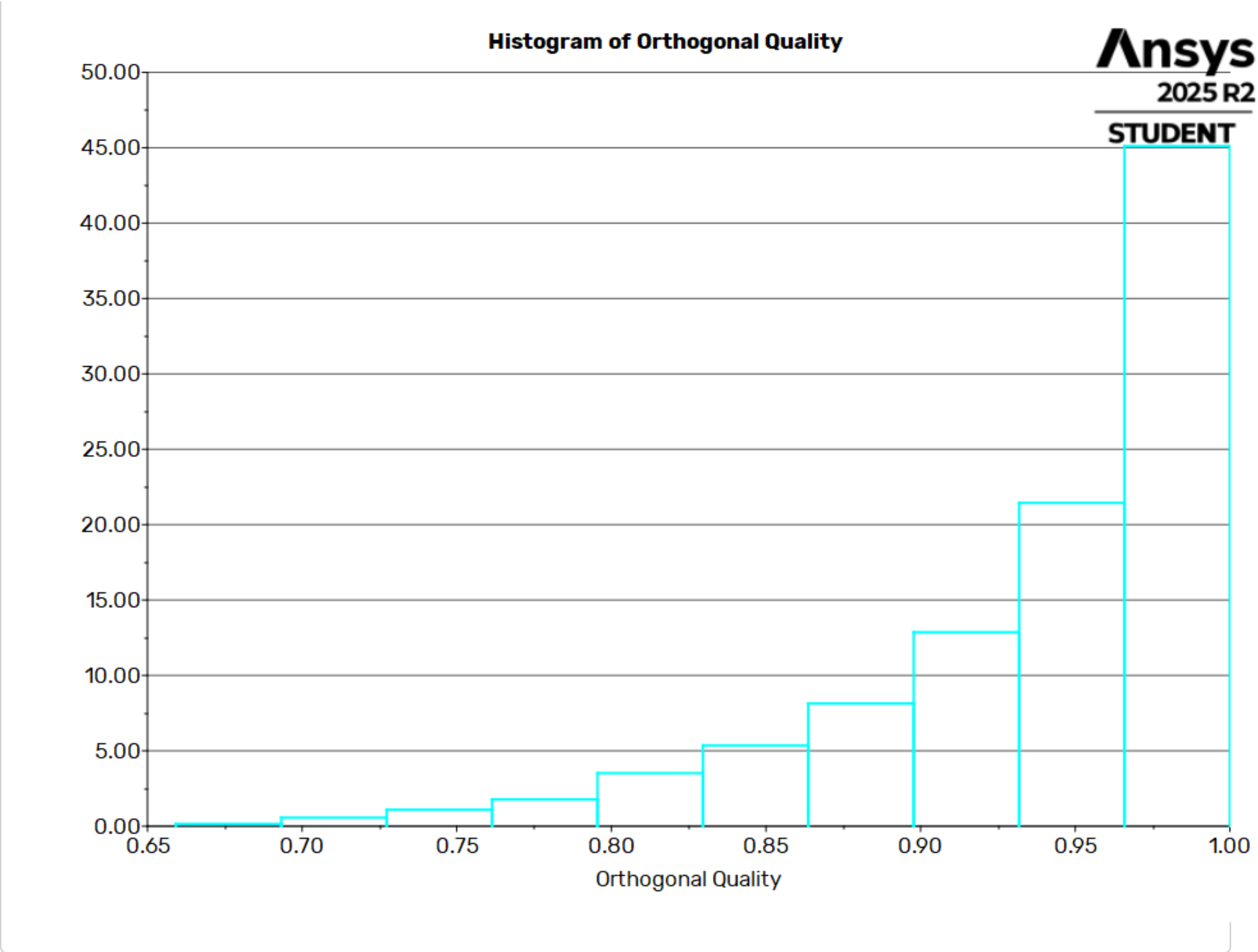
Mesh Size

Cells	Faces	Nodes
39	140	0
118	415	0
351	1192	0
1023	3254	0
3288	9800	0
22914	35063	12153

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fluid	Tri Cell	0.65988415	3.7977613
solid	Tri Cell	0.65917937	3.7967206

Orthogonal Quality


Histogram of Orthogonal Quality
Ansys
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STUDENT
50.00
45.00
40.00
35.00
30.00
25.00
20.00
15.00
10.00
5.00
0.00
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Orthogonal Quality

Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Steady
Viscous	Laminar
Heat Transfer	Enabled

Material Properties

− Fluid	
− air	
Density	ideal gas
Cp (Specific Heat)	1006.43 J/(kg K)
Thermal Conductivity	0.0242 W/(m K)
Viscosity	1.7894e-05 kg/(m s)
Molecular Weight	28.966 kg/kmol
− Solid	
− aluminum	
Density	2719 kg/m^3
Cp (Specific Heat)	871 J/(kg K)
Thermal Conductivity	202.4 W/(m K)

Cell Zone Conditions

− Fluid	
− fluid	
Material Name	air

Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Porous zone?	no
– Solid	
– solid	
Material Name	aluminum
Specify source terms?	no
Frame Motion?	no
Solid Motion?	no

Boundary Conditions

– Inlet	
– inlet	
Gauge Pressure [Pa]	101325
Mach Number	1.1
Temperature [K]	300
Component of Flow Direction (x,y)	(1, 0)
– outlet	
Gauge Pressure [Pa]	101325
Mach Number	1.1
Temperature [K]	300
Component of Flow Direction (x,y)	(1, 0)
– Wall	
– soild_wall	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Convective Augmentation Factor	1

− soild_wall:001	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Coupled
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Convective Augmentation Factor	1
− soild_wall:001-shadow	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Coupled
Convective Augmentation Factor	1

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Depth	1 m
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	fluid

Solver Settings

− Equations	
Flow	True

− Numerics	
Absolute Velocity Formulation	True
− Under-Relaxation Factors	
Solid	1
− Discretization Scheme	
Flow	Second Order Upwind
− Time Marching	
Solver	Explicit
Courant Number	0.05
Residual Smoothing Iterations	0 (disabled)
Residual Smoothing Factor	0.5
− Multigrid Solver	
Coarse Grid Levels	5
Fine Grid Iterations	1
Final Iterations	0
Time Step Reduction	0.9
Correction Reduction	0.6
Species Correction Reduction	1
Correction Smoothing Factor	0.5
− Multi-Stage Scheme	
1	0.2075
2	0.5915
3	1
− Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000

Run Information

Number of Machines	1
Number of Cores	2
Case Read	7.168 seconds
Iteration	142.937 seconds
Virtual Current Memory	1.07743 GB
Virtual Peak Memory	1.13433 GB
Memory Per M Cell	18.6047

Solution Status

Iterations: 5473

	Value	Absolute Criteria	Convergence Status
continuity	0.0009993356	0.001	Converged
x-velocity	0.0001674192	0.001	Converged
y-velocity	0.0003978381	0.001	Converged
energy	0.0008814828	0.001	Converged

Named Expressions

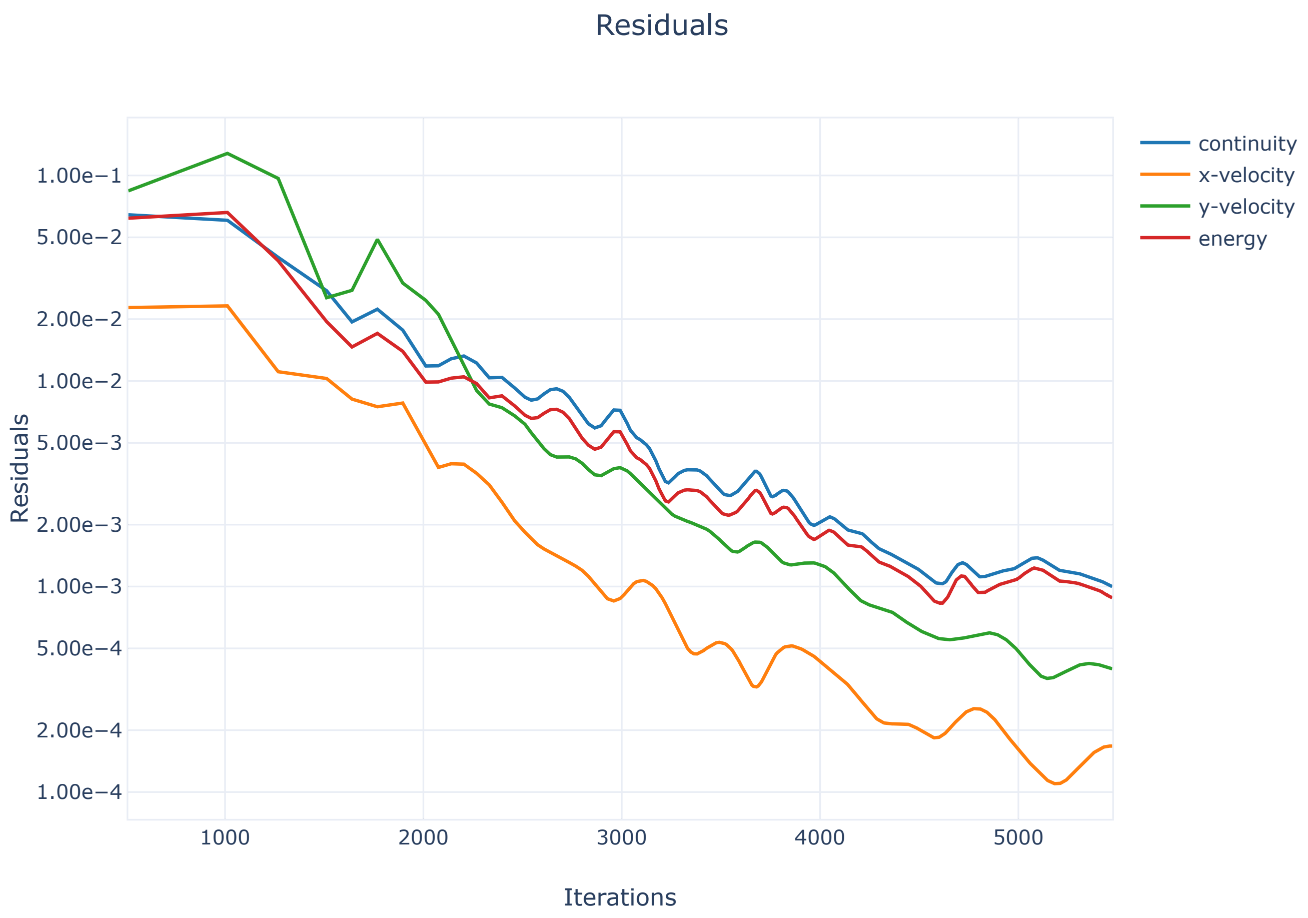
Expression	Definition	Value	Unit	Used In	Description
expr1	Maximum(MachNumber, ['fluid'])	2.0261369	[]	report-def-0 (Report Definition)	

Report Definitions

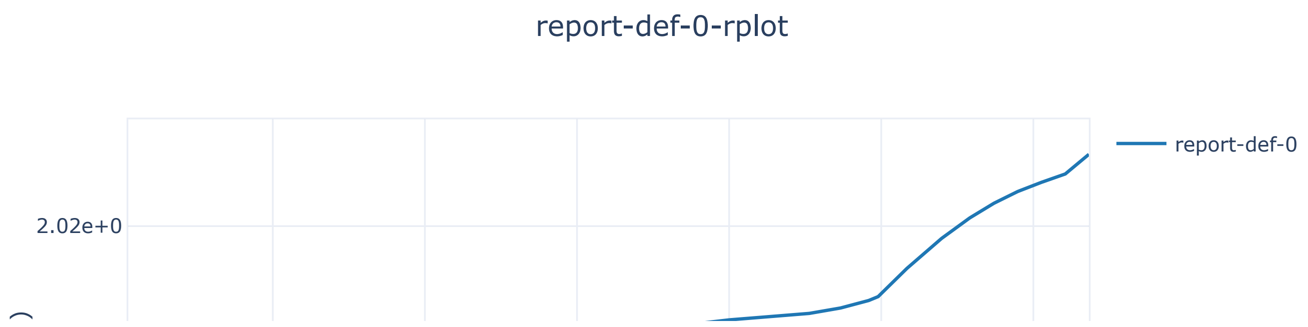
report-def-0	2.026137

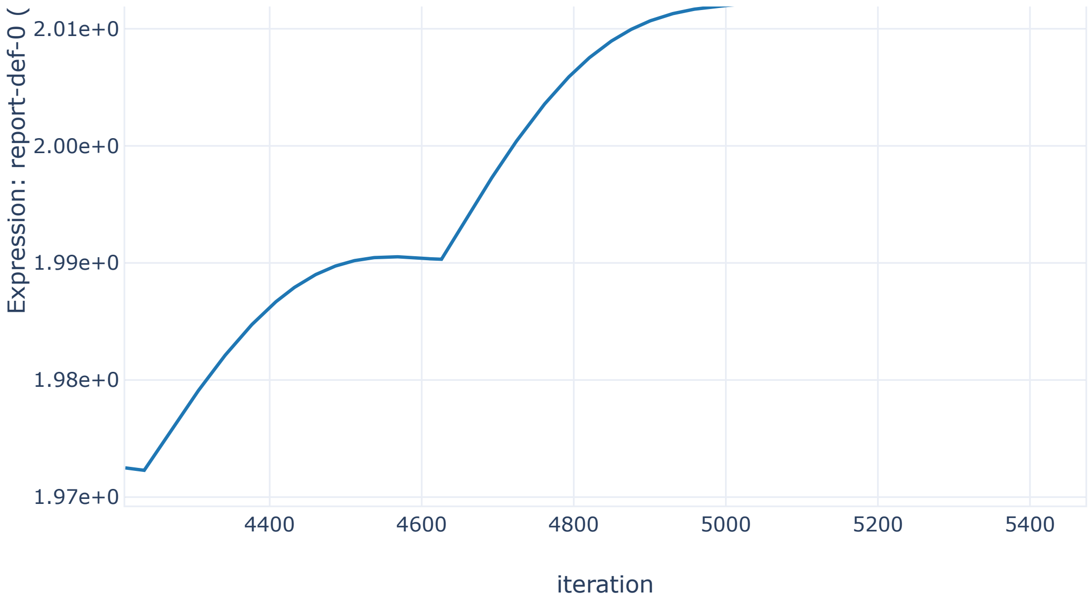
Plots

Residuals



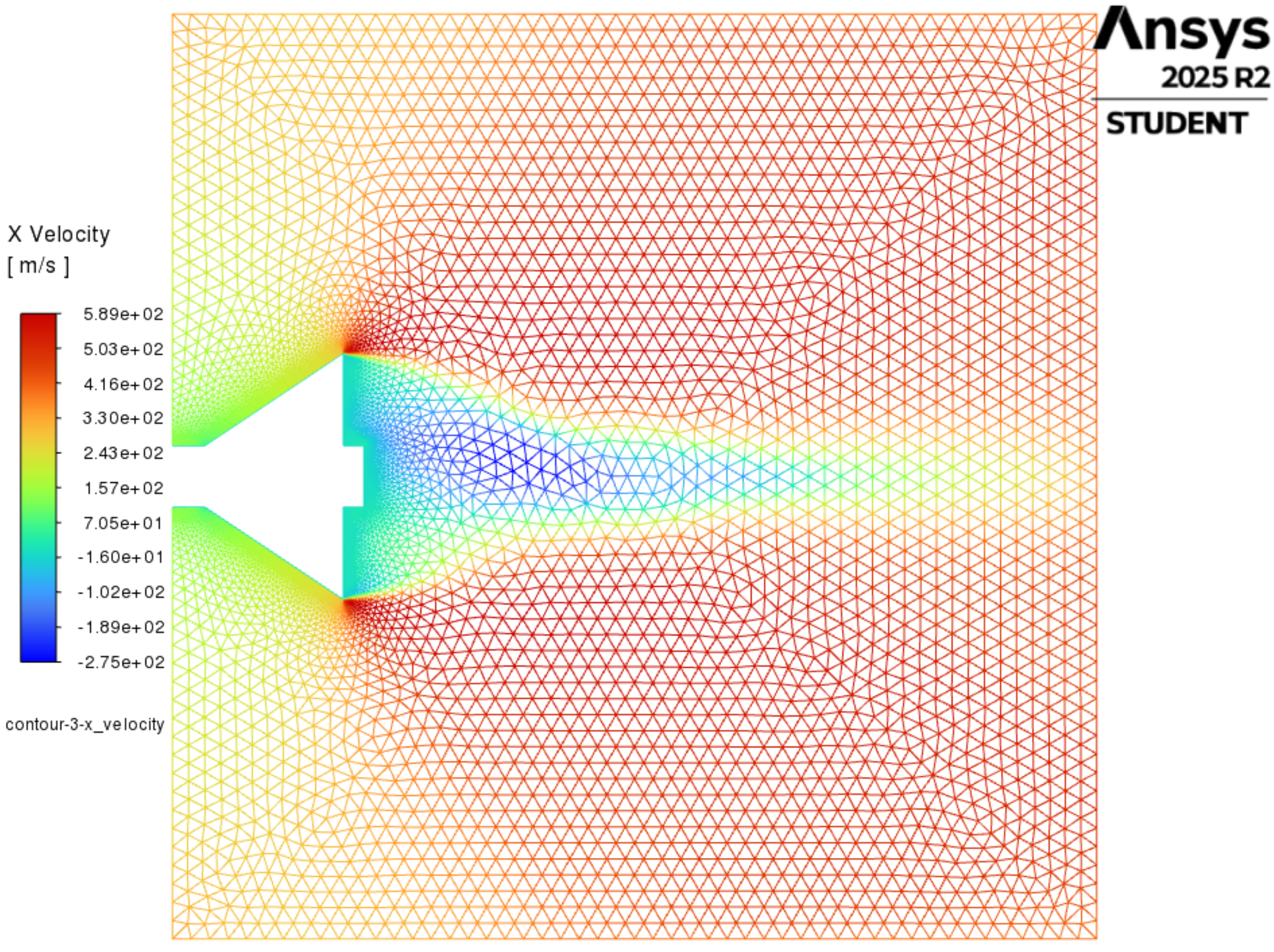
report-def-0-rplot



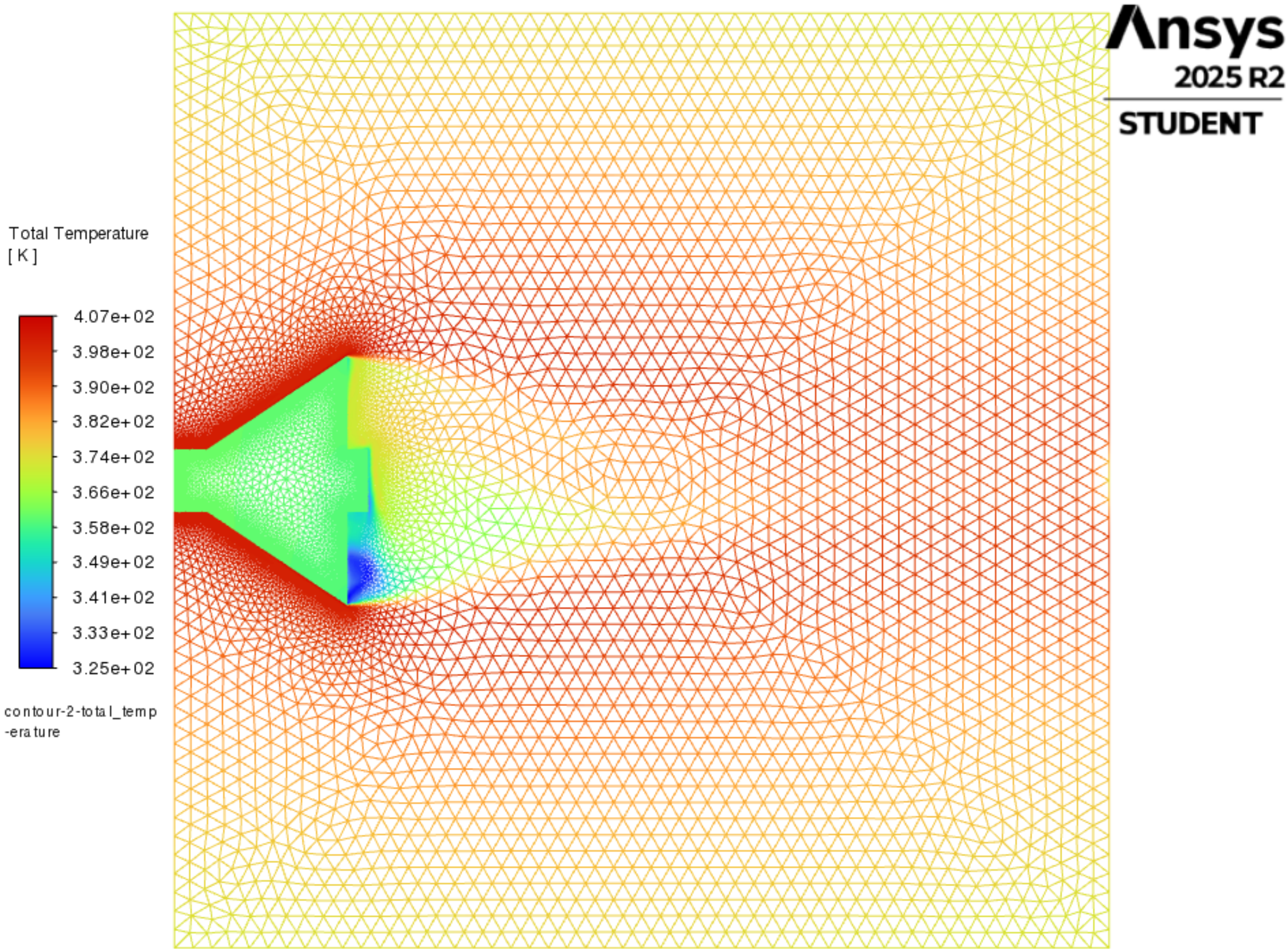


Contours

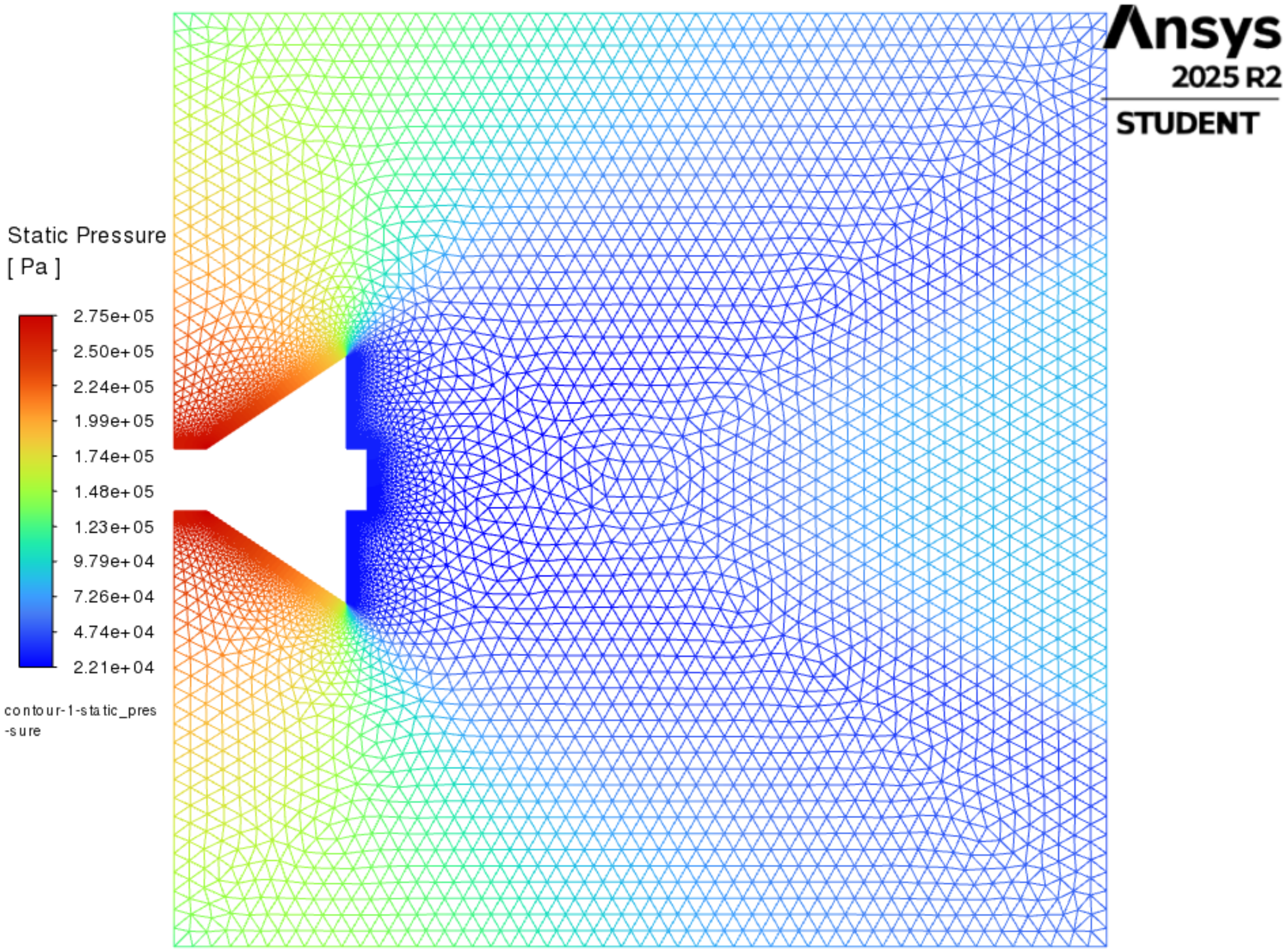
contour-3-x_velocity



contour-2-total_temperature



contour-1-static_pressure


Static Pressure
[Pa]
2.75e+05
2.50e+05
2.24e+05
1.99e+05
1.74e+05
1.48e+05
1.23e+05
9.79e+04
7.26e+04
4.74e+04
2.21e+04
contour-1-static_pres-sure
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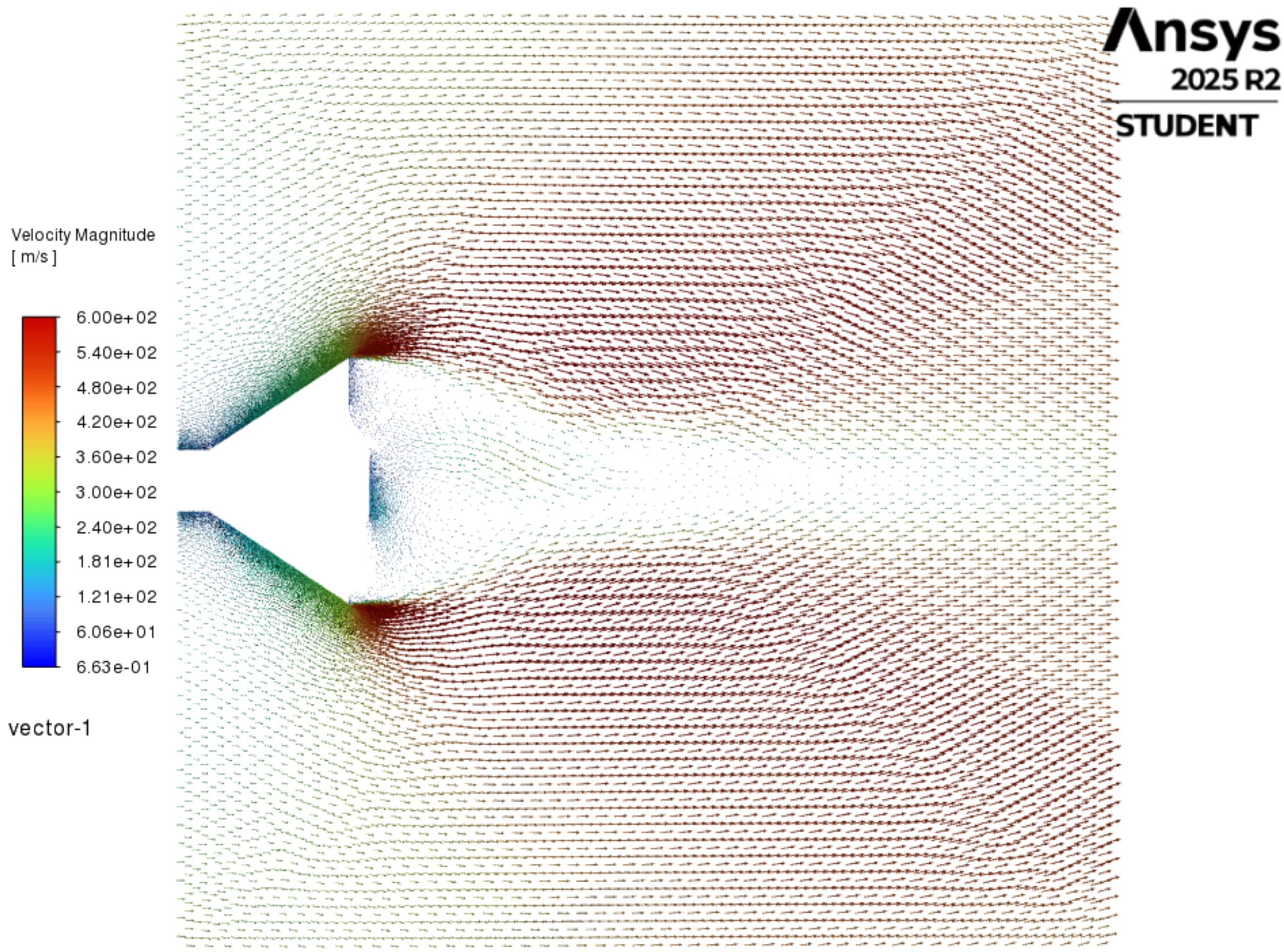
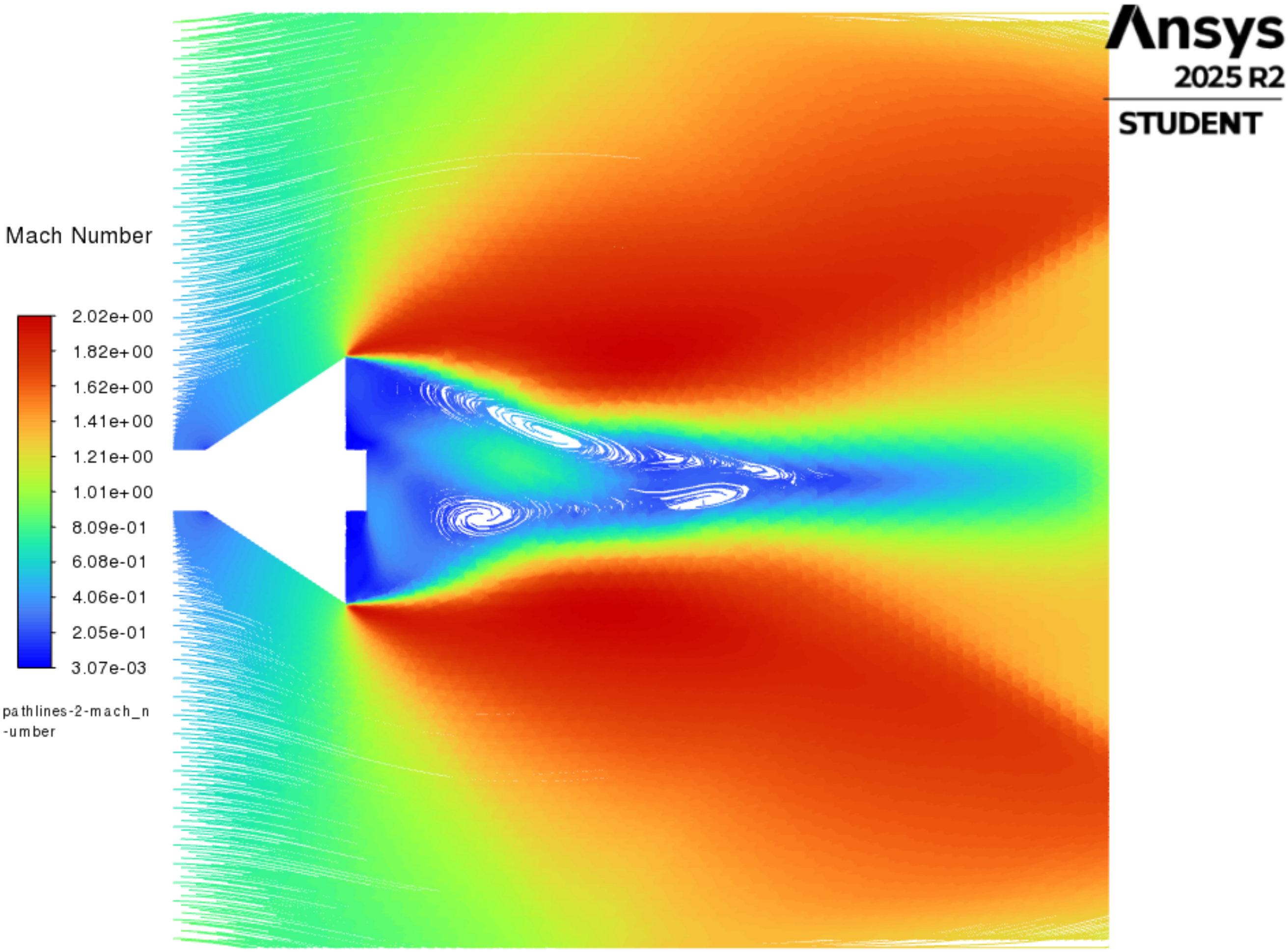
Vectors

vector-1

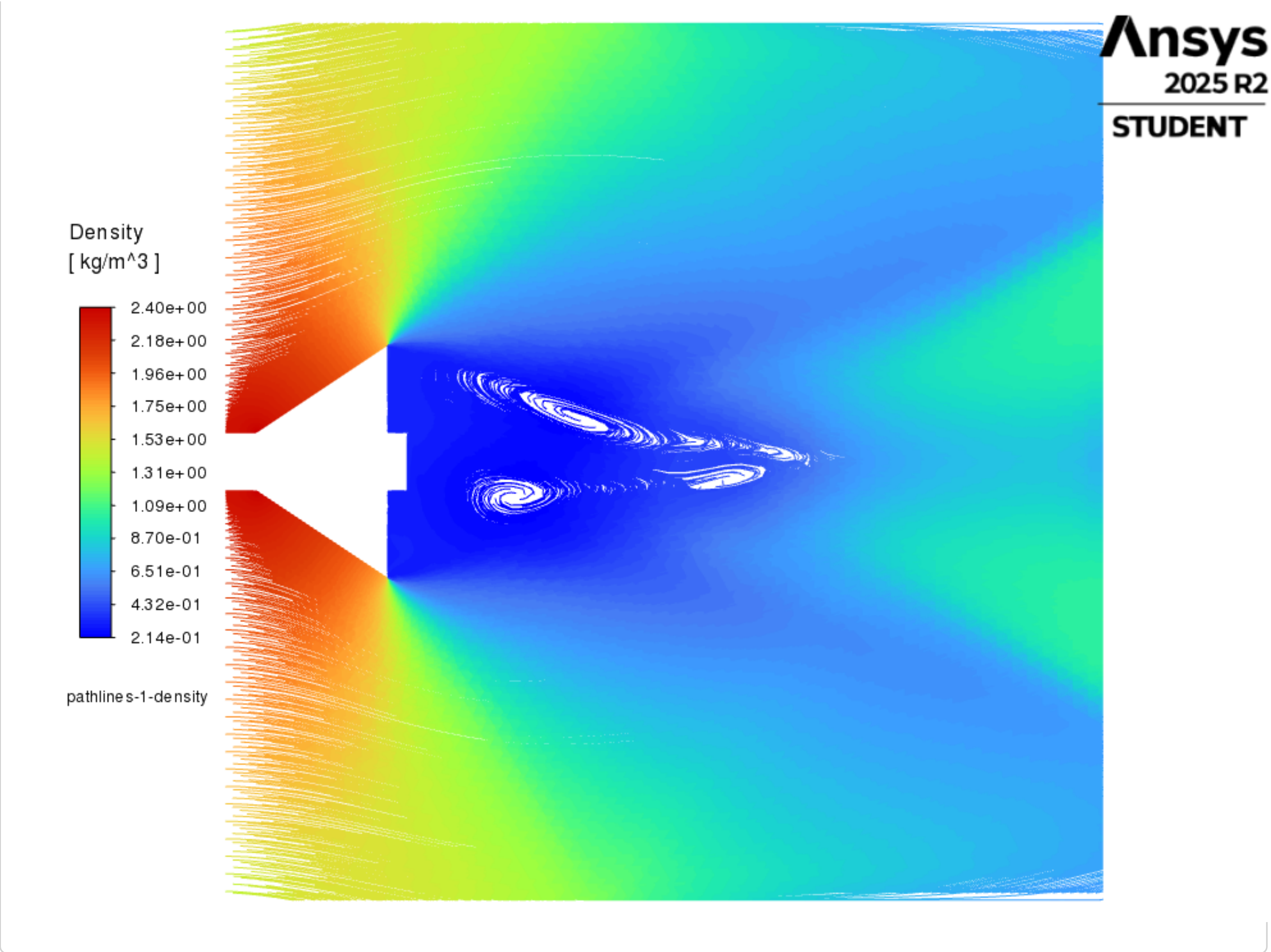


Pathlines

pathlines-2-mach_number



pathlines-1-density


Density
[kg/m^3]
2.40e+00
2.18e+00
1.96e+00
1.75e+00
1.53e+00
1.31e+00
1.09e+00
8.70e-01
6.51e-01
4.32e-01
2.14e-01
pathlines-1-density
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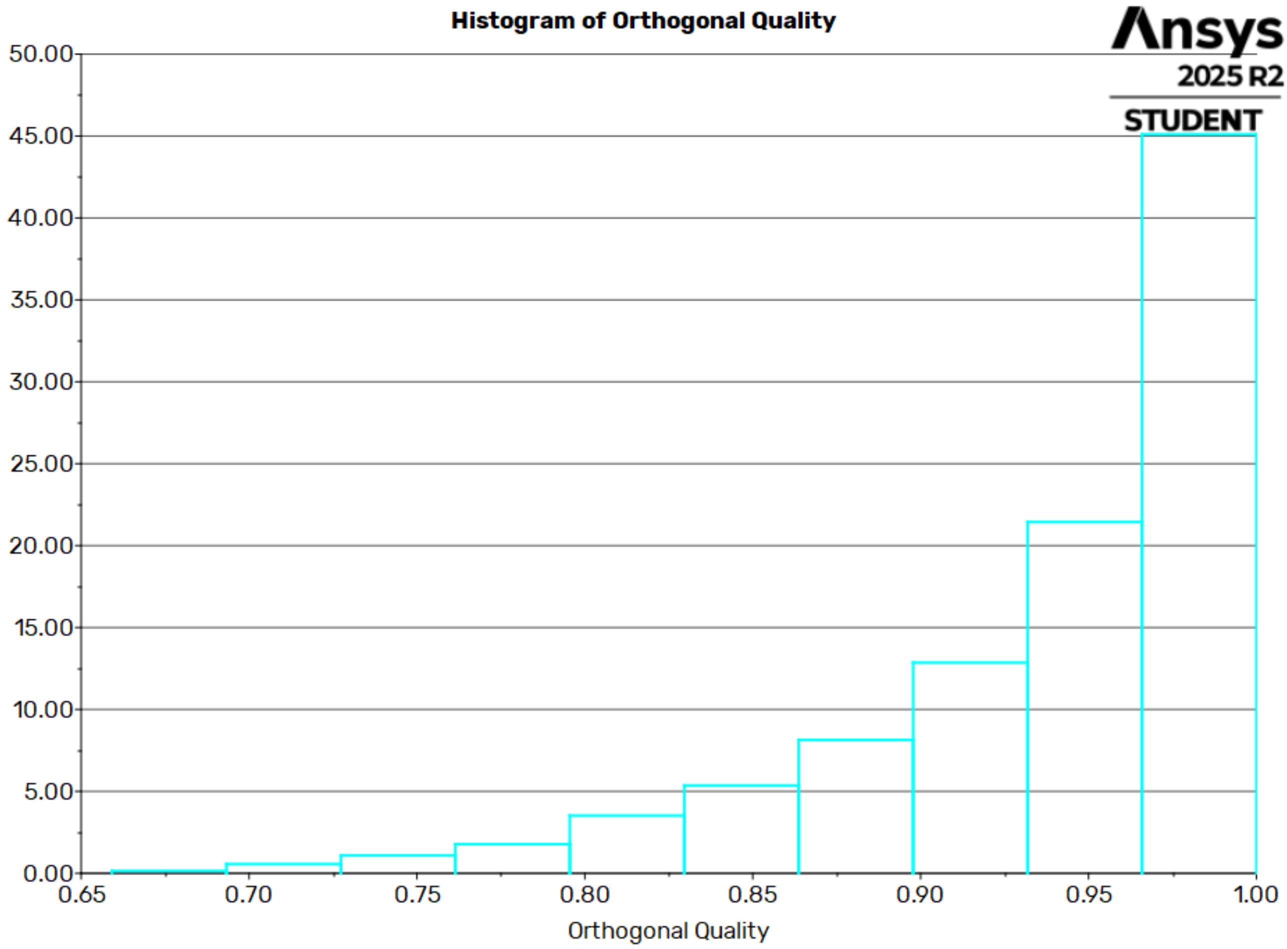
Scenes


Histogram of Orthogonal Quality
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50.00
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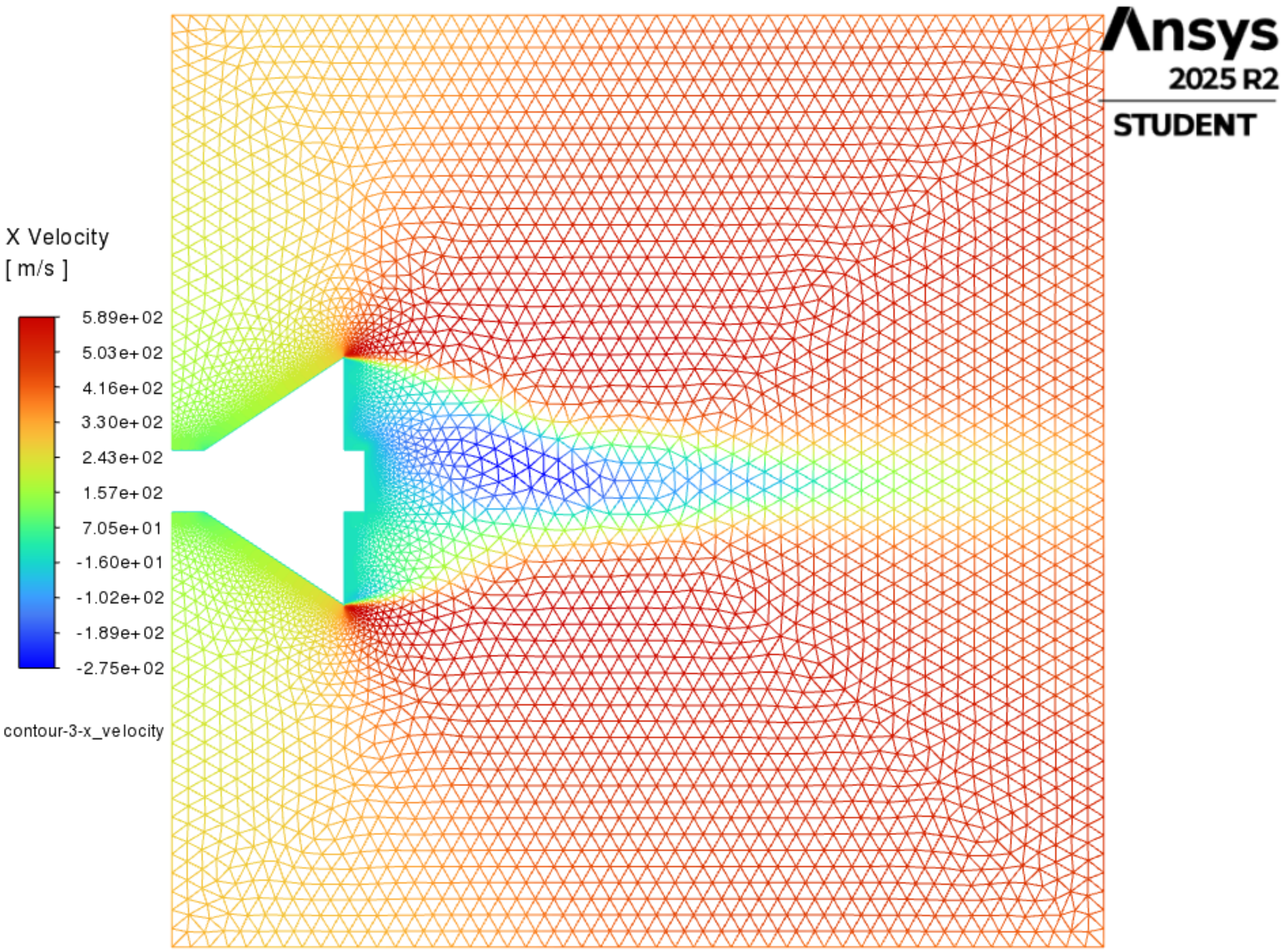
40.00
35.00
30.00
25.00
20.00
15.00
10.00
5.00
0.00
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Orthogonal Quality


Ansys
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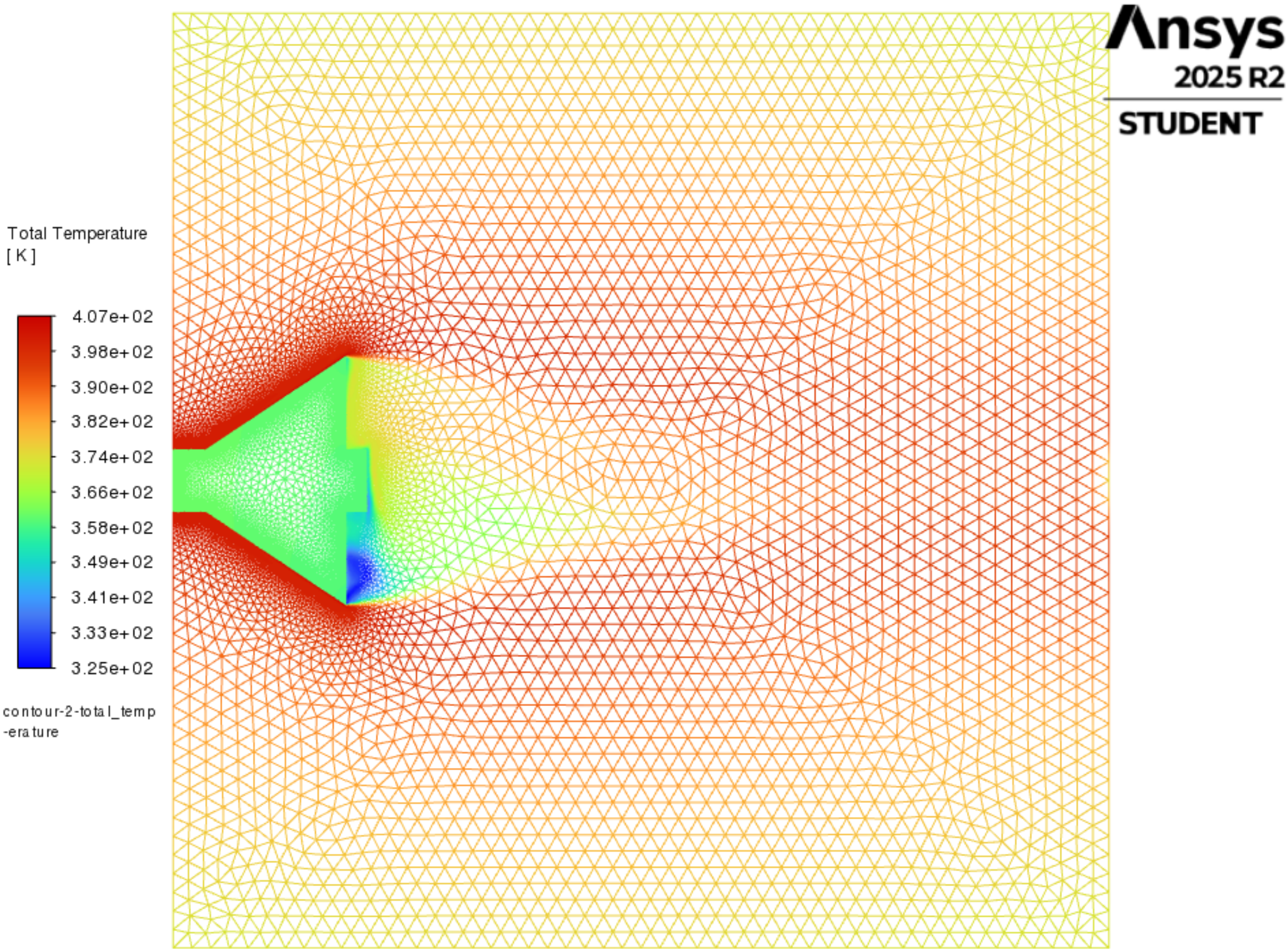
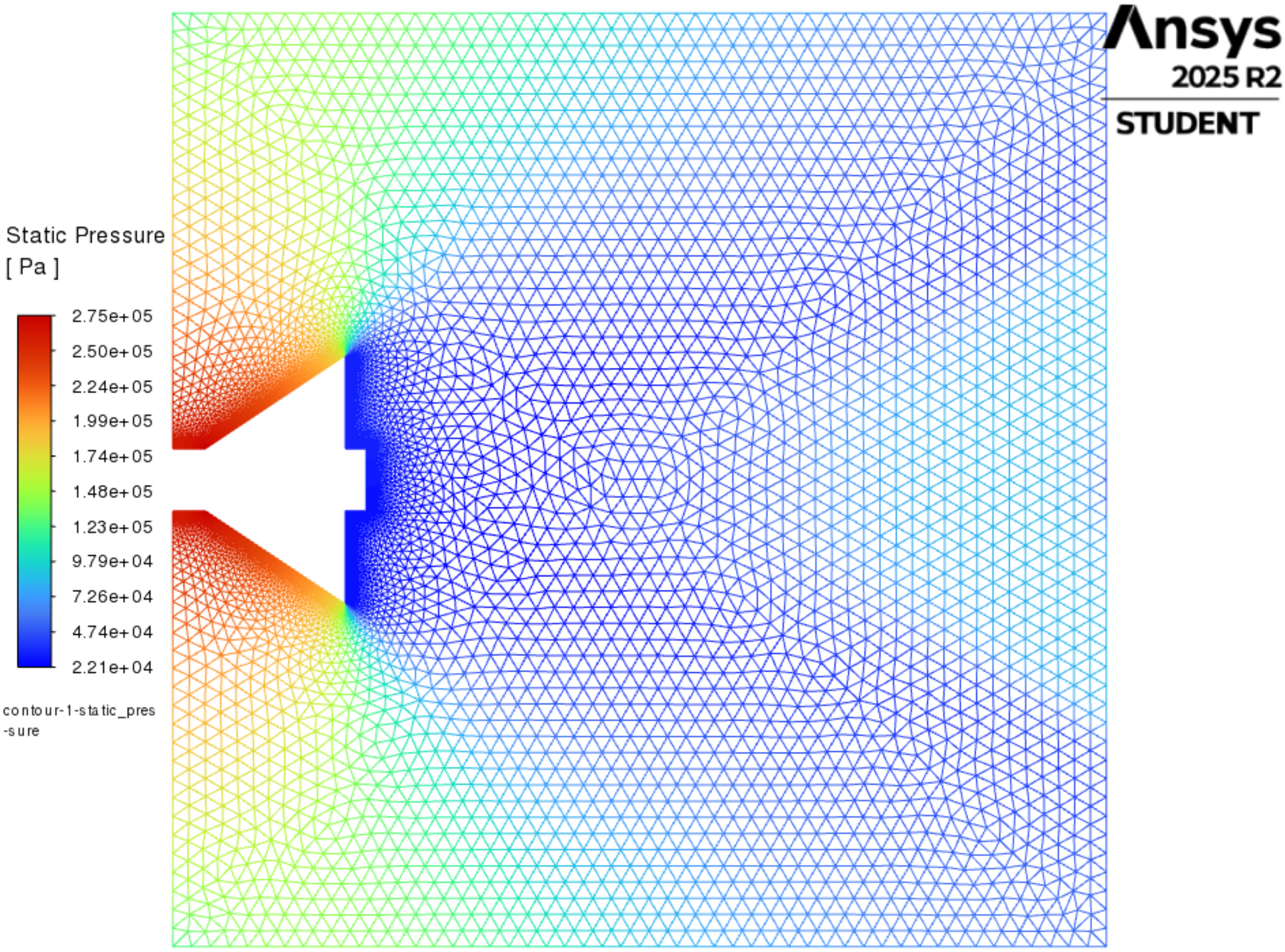
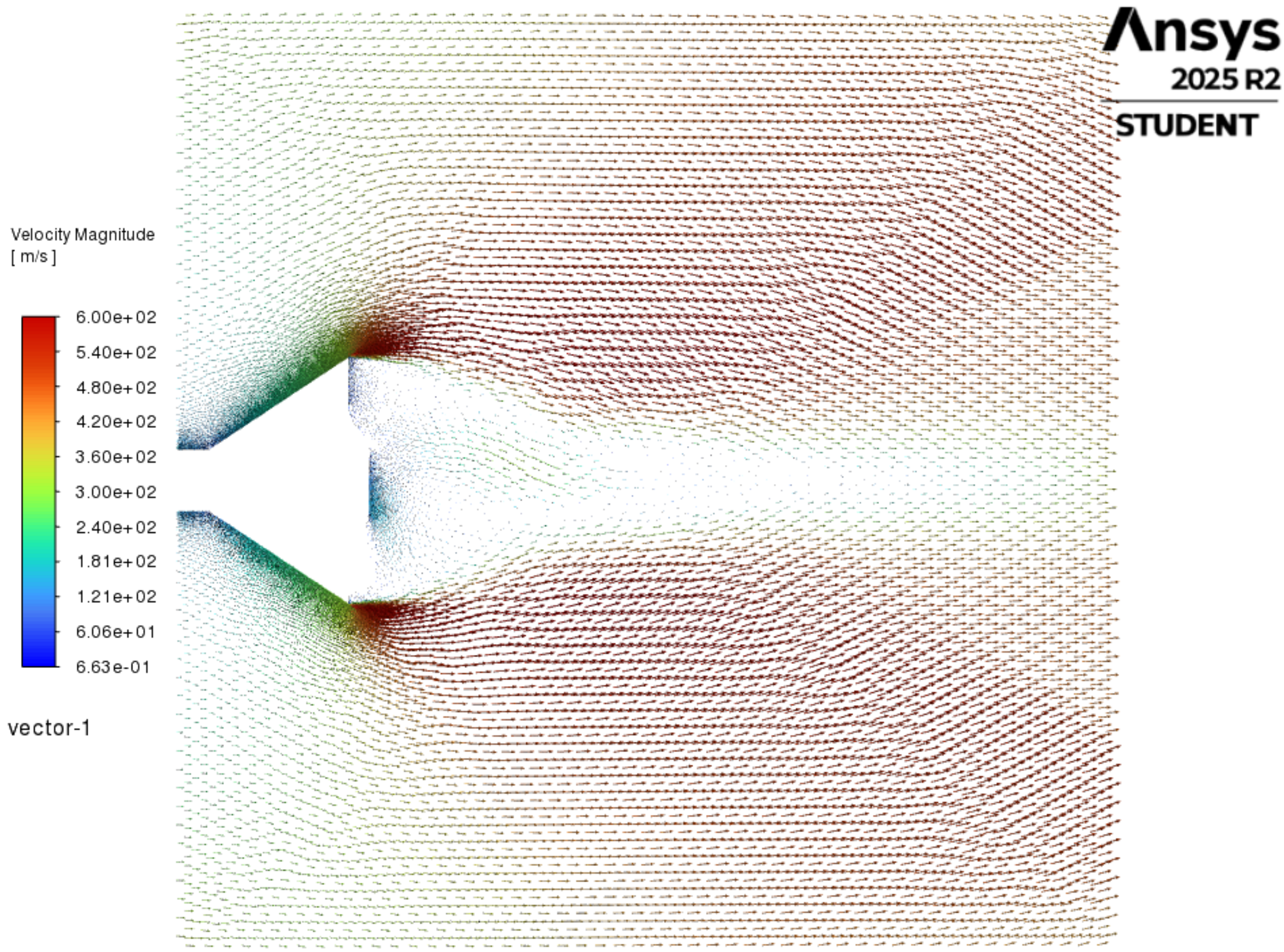
STUDENT
X Velocity
[m/s]
5.89e+02
5.03e+02
4.16e+02
3.30e+02
2.43e+02
1.57e+02
7.05e+01
-1.60e+01
-1.02e+02
-1.89e+02
-2.75e+02
contour-3-x_velocity


Ansys
2025 R2
STUDENT
Total Temperature
[K]
4.07e+02
3.98e+02
3.90e+02
3.82e+02
3.74e+02
3.66e+02
3.58e+02
3.49e+02
3.41e+02
3.33e+02
3.25e+02
contour-2-total_temp
-erature


Ansys
2025 R2
STUDENT
Static Pressure
[Pa]
2.75e+05
2.50e+05
2.24e+05
1.99e+05
1.74e+05
1.48e+05
1.23e+05
9.79e+04
7.26e+04
4.74e+04
2.21e+04
contour-1-static_pres
-sure


Ansys
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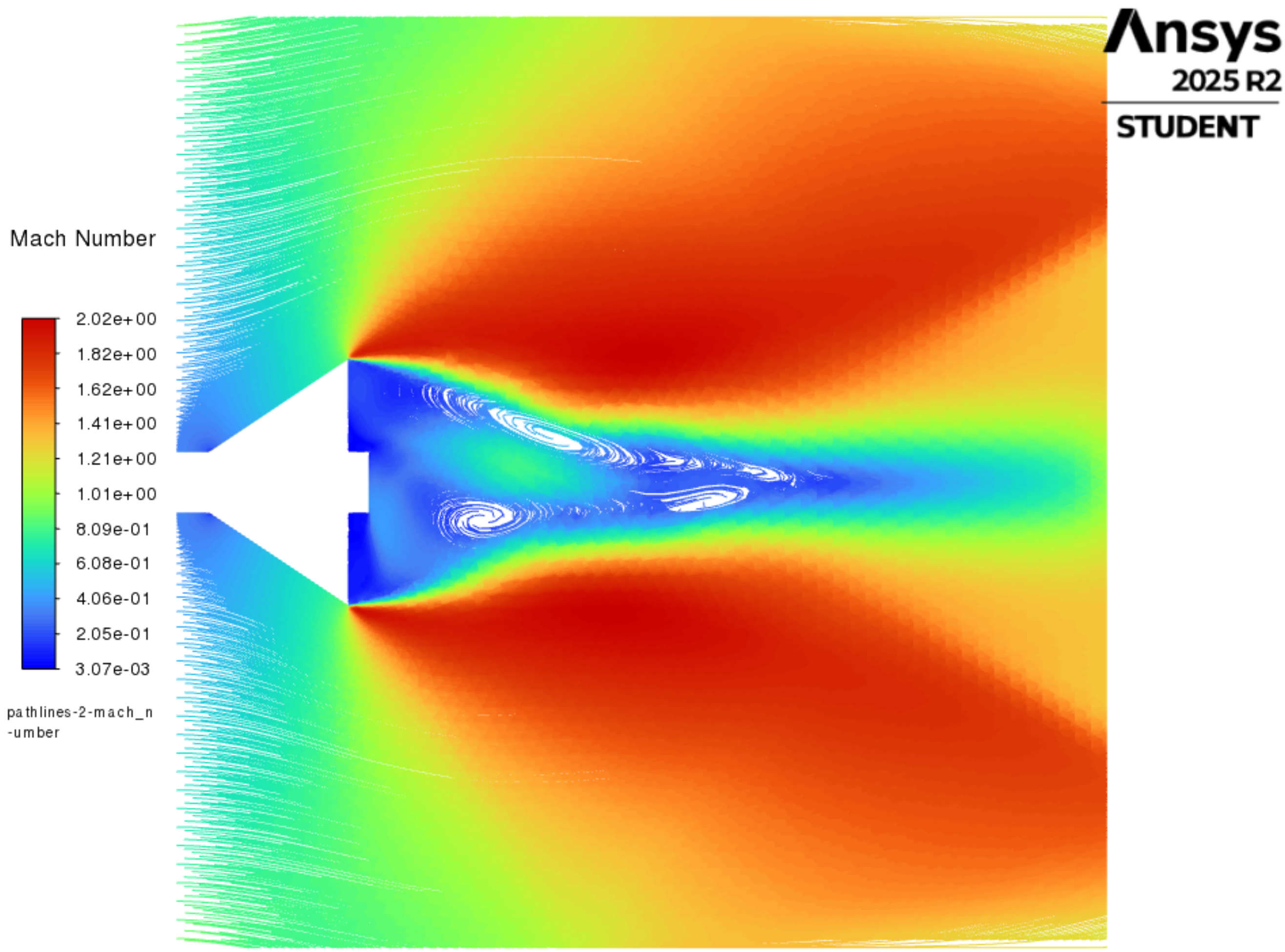
Velocity Magnitude
[m/s]
6.00e+02
5.40e+02
4.80e+02
4.20e+02
3.60e+02
3.00e+02
2.40e+02
1.81e+02
1.21e+02
6.06e+01
6.63e-01
vector-1


Ansys
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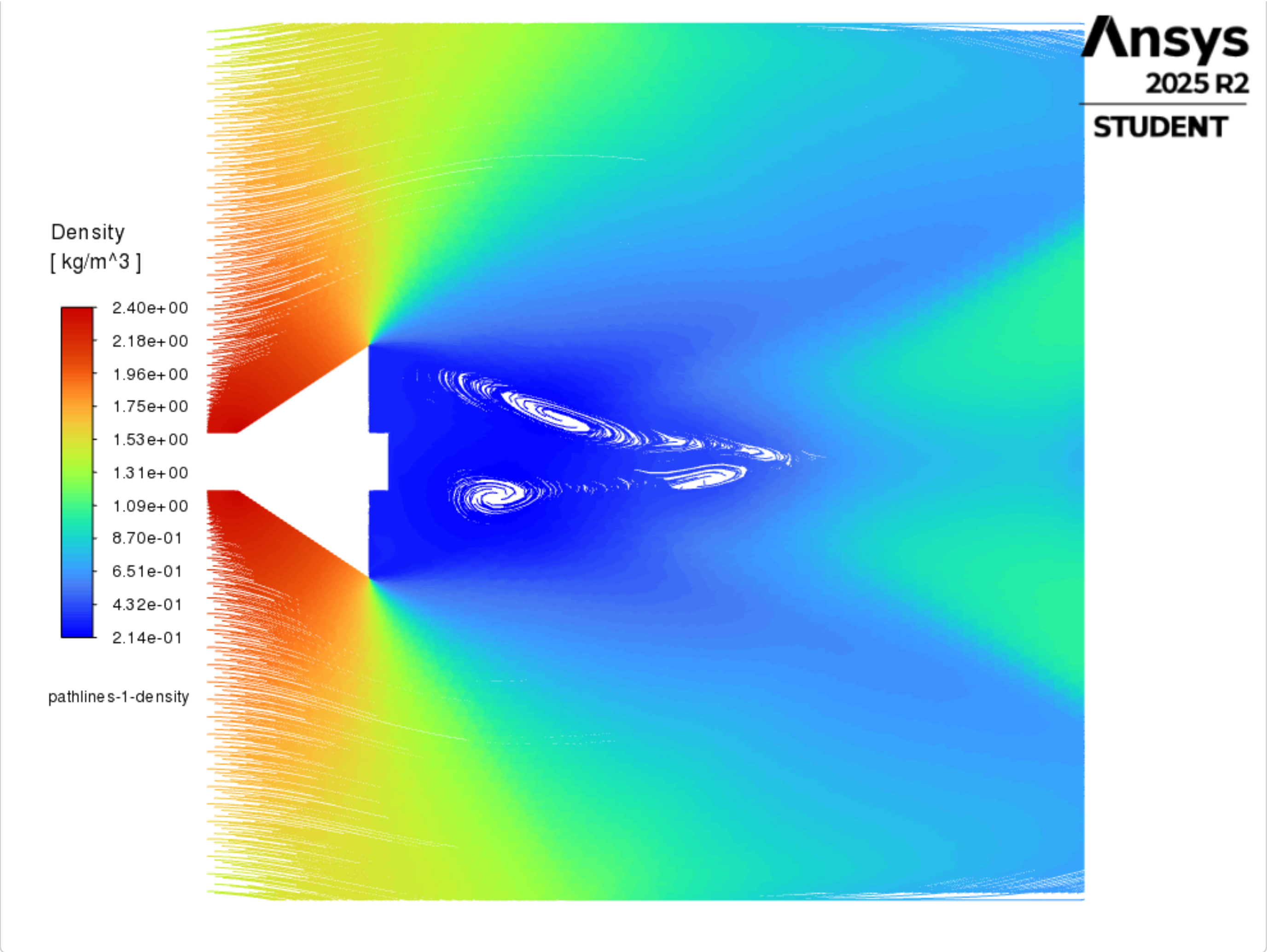
STUDENT
Mach Number
2.02e+00
1.82e+00
1.62e+00
1.41e+00
1.21e+00
1.01e+00
8.09e-01
6.08e-01
4.06e-01
2.05e-01
3.07e-03
pathlines-2-mach_n
-umber



Ansys
2025 R2
STUDENT
Density
[kg/m^3]
2.40e+00
2.18e+00
1.96e+00
1.75e+00
1.53e+00
1.31e+00
1.09e+00
8.70e-01
6.51e-01
4.32e-01
2.14e-01
pathlines-1-density